

# CCNA Subnetting Drill Sheet

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## 50+ Problems with Full Worked Answers — From Baby Steps to Exam Speed

**Why this exists:** Subnetting shows up ALL OVER the CCNA exam, and you have to do it **fast** (about 60–90 seconds per question). Reading about it isn't enough — you have to drill it until it's automatic, like times tables. This sheet gives you 50+ problems, grouped from easy to hard, each with a **step-by-step answer**.

**How to use it:** Cover the answer with your hand or a piece of paper. Work the problem on scratch paper **FIRST**. Then check. Do **5–10 problems every single day**. In two weeks you'll be fast.

**Where this fits.** The theory behind every method here lives in **Chapters 11–13 of the CCNA Study Guide** (IP addressing, subnetting, IPv6) — if a step doesn't make sense, that's where to look. Once these are automatic, the **Practice Question Bank** puts subnetting back into full exam-style questions.

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## The Toolkit (Everything You Need on One Page)

### The Two Magic Charts

#### Chart 1 — Bit values in one octet (memorize left to right):

|     |    |    |    |   |   |   |   |
|-----|----|----|----|---|---|---|---|
| 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1 |
|-----|----|----|----|---|---|---|---|

#### Chart 2 — The subnet "cheat table" (memorize this cold):

| CIDR | Mask (last octet) | Block Size | Hosts (usable) |
|------|-------------------|------------|----------------|
| /24  | 0                 | 256        | 254            |
| /25  | 128               | 128        | 126            |
| /26  | 192               | 64         | 62             |
| /27  | 224               | 32         | 30             |
| /28  | 240               | 16         | 14             |
| /29  | 248               | 8          | 6              |
| /30  | 252               | 4          | 2              |

Notice the pattern: **Block size doubles as you go up** (4, 8, 16, 32, 64, 128, 256), and **Mask value** = 256 – block size.

### The 5 Facts You Compute for Any Subnet

For any IP + mask, you can always find these 5 things:

1. **Block size** = 256 – (the interesting octet of the mask).
2. **Network address** = the block start the IP falls into (all host bits 0).
3. **Broadcast address** = one below the next network (all host bits 1).
4. **First usable host** = network + 1.
5. **Last usable host** = broadcast – 1.

## The Formulas

```
Number of hosts per subnet = 2^(host bits) - 2
Number of subnets         = 2^(borrowed bits)
Host bits                  = 32 - CIDR
```

## The 3-Step "Where Does This IP Live?" Method

1. Find the **block size** (256 – interesting mask octet).
2. Count up in the interesting octet by the block size (0, block, 2×block...) until you pass the IP's value. The one **just below or equal** is the **network**.
3. **Broadcast** = next network – 1. Hosts sit between them.

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# SECTION 1 — Warm-Up: Masks, CIDR & Host Counts

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(Goal: instant recall. Aim for under 15 seconds each.)

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**Q1.** Convert **/26** to a dotted-decimal mask.

**Answer**

**255.255.255.192**

/26 means 26 ones: **11111111.11111111.11111111.11000000**. The last octet **11000000** = 128 + 64 = **192**.

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**Q2.** Convert **255.255.255.240** to CIDR.

**Answer**

**/28**

240 = 11110000 = 4 ones. So  $8 + 8 + 8 + 4 = 28$ .

**Q3.** Convert /19 to a dotted-decimal mask.

**Answer**

**255.255.224.0**

/19 = 11111111.11111111.11100000.00000000. Third octet 11100000 =  $128 + 64 + 32 = 224$ .

**Q4.** How many **usable hosts** are in a /27?

**Answer**

**30**

Host bits =  $32 - 27 = 5$ . Hosts =  $2^5 - 2 = 32 - 2 = 30$ .

**Q5.** How many **usable hosts** are in a /30?

**Answer**

**2**

Host bits =  $32 - 30 = 2$ . Hosts =  $2^2 - 2 = 4 - 2 = 2$ . (This is the classic router-to-router link size.)

**Q6.** How many **usable hosts** are in a /22?

**Answer**

**1022**

Host bits =  $32 - 22 = 10$ . Hosts =  $2^{10} - 2 = 1024 - 2 = 1022$ .

**Q7.** Convert binary 11000000 to decimal.

**Answer**

**192**

$128 + 64 = 192$ .

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**Q8.** Convert decimal 224 to binary.

**Answer**

**11100000**

$128 + 64 + 32 = 224$ . So the top three bits are on: 11100000.

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**Q9.** What is the **block size** of a /29?

**Answer**

**8**

/29 mask last octet = 248. Block size =  $256 - 248 = 8$ .

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**Q10.** A mask ends in .248. What CIDR is that, and how many hosts?

**Answer**

**/29, 6 hosts**

$248 = 11111000 = 5 \text{ ones} \rightarrow 24 + 5 = /29$ . Host bits = 3  $\rightarrow 2^3 - 2 = 6$ .

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## SECTION 2 — Find the Network & Broadcast Address

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(The bread-and-butter skill. Aim for under 60 seconds each.)

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**Q11.** Given 192.168.1.77 /26, find the network and broadcast address.

**Answer**

**Network 192.168.1.64, Broadcast 192.168.1.127**

- Block size =  $256 - 192 = 64$ .
- Count by 64: 0, 64, 128... 77 falls between **64 and 127**.
- Network = **.64**, next network = .128, so Broadcast =  $128 - 1 = .127$ .

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**Q12.** Given 10.0.0.200 /27, find the network and broadcast.

**Answer**

**Network 10.0.0.192, Broadcast 10.0.0.223**

- Block size =  $256 - 224 = 32$ .
- Count by 32: ...160, 192, 224. 200 is between **192 and 223**.
- Network = **.192**, Broadcast =  $224 - 1 = .223$ .

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**Q13.** Given 172.16.5.130 /25, find the network and broadcast.

**Answer**

**Network 172.16.5.128, Broadcast 172.16.5.255**

- Block size =  $256 - 128 = 128$ .
- Count by 128: 0, 128. 130 is between **128 and 255**.
- Network = **.128**, Broadcast = **.255**.

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**Q14.** Given 192.168.10.44 /28, find the network and broadcast.

**Answer**

**Network 192.168.10.32, Broadcast 192.168.10.47**

- Block size =  $256 - 240 = 16$ .
- Count by 16: 0, 16, 32, 48. 44 is between **32 and 47**.
- Network = **.32**, Broadcast =  $48 - 1 = .47$ .

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**Q15.** Given 10.10.10.10 /29, find the network and broadcast.

**Answer**

### Network 10.10.10.8, Broadcast 10.10.10.15

- Block size =  $256 - 248 = 8$ .
- Count by 8: 0, 8, 16. 10 is between **8 and 15**.
- Network = **.8**, Broadcast =  $16 - 1 = .15$ .

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**Q16.** Given 192.168.1.100 /30, find the network and broadcast.

### Answer

### Network 192.168.1.100, Broadcast 192.168.1.103

- Block size =  $256 - 252 = 4$ .
- Count by 4: ...96, 100, 104. 100 lands exactly ON **100** (a network boundary).
- Network = **.100**, Broadcast =  $104 - 1 = .103$ .

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**Q17.** Given 172.16.99.200 /21, find the network and broadcast. (Now the interesting octet is the THIRD one!)

### Answer

### Network 172.16.96.0, Broadcast 172.16.103.255

- /21 mask = 255.255.**248**.0. The interesting octet is the **3rd**.
- Block size =  $256 - 248 = 8$  (in the third octet).
- Count by 8 in octet 3: ...88, 96, 104. 99 is between **96 and 103**.
- Network = 172.16.**96**.0, Broadcast = 172.16.**103**.255.

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**Q18.** Given 10.50.150.7 /20, find the network and broadcast.

### Answer

### Network 10.50.144.0, Broadcast 10.50.159.255

- /20 mask = 255.255.**240**.0. Interesting octet = 3rd.

- Block size =  $256 - 240 = 16$ .
- Count by 16 in octet 3: ...128, 144, 160. 150 is between **144 and 159**.
- Network = 10.50.**144**.0, Broadcast = 10.50.**159**.255.

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**Q19.** Given **192.168.200.55 /18**, find the network and broadcast.

**Answer**

**Network 192.168.192.0, Broadcast 192.168.255.255**

- /18 mask = 255.255.**192**.0. Interesting octet = 3rd.
- Block size =  $256 - 192 = 64$ .
- Count by 64 in octet 3: 0, 64, 128, 192, 256. 200 is between **192 and 255**.
- Network = 192.168.**192**.0, Broadcast = 192.168.**255**.255.

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**Q20.** Given **10.1.1.1 /8**, find the network and broadcast.

**Answer**

**Network 10.0.0.0, Broadcast 10.255.255.255**

- /8 mask = 255.0.0.0. The whole 10.x.x.x block is one network.
- Network = **10.0.0.0**, Broadcast = **10.255.255.255**.

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## SECTION 3 — Find the Usable Host Range

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(Network + 1 to Broadcast – 1. Put it all together.)

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**Q21.** For **192.168.1.77 /26**, list the **first and last usable host**.

**Answer**

**First 192.168.1.65, Last 192.168.1.126**

Network = .64, Broadcast = .127 (from Q11). First host =  $64 + 1 = \mathbf{.65}$ , Last host =  $127 - 1 = \mathbf{.126}$ .

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**Q22.** For **172.16.5.130 /25**, list the first and last usable host.

**Answer**

**First 172.16.5.129, Last 172.16.5.254**

Network = .128, Broadcast = .255. First = **.129**, Last = **.254**.

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**Q23.** For **10.10.10.10 /29**, list the first and last usable host.

**Answer**

**First 10.10.10.9, Last 10.10.10.14**

Network = .8, Broadcast = .15. First = **.9**, Last = **.14**.

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**Q24.** For **192.168.1.100 /30**, list the first and last usable host.

**Answer**

**First 192.168.1.101, Last 192.168.1.102**

Network = .100, Broadcast = .103. First = **.101**, Last = **.102**. (Exactly 2 usable — perfect for a WAN link.)

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**Q25.** For **172.16.99.200 /21**, list the first and last usable host.

**Answer**

**First 172.16.96.1, Last 172.16.103.254**

Network = 172.16.96.0, Broadcast = 172.16.103.255. First = 172.16.96.**1**, Last = 172.16.103.**254**.

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**Q26.** Is **192.168.1.64** a **usable host** address in the **192.168.1.0 /26** scheme? Why?

**Answer**

**No — it's a network address.**



With /26, block size 64, the subnets start at .0, .64, .128, .192. **.64** is the **network address** of the second subnet, so it can't be assigned to a device.

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**Q27.** Is 192.168.1.191 usable in a /26 scheme? Why?

**Answer**

**No — it's a broadcast address.**

Subnets: .128–.191 is the third /26 (block .128, next block .192). So **.191** is the **broadcast** of that subnet — not usable.

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**Q28.** Two PCs: 192.168.1.70 /26 and 192.168.1.130 /26. Can they talk **directly** (same subnet)?

**Answer**

**No — different subnets, so they need a router.**

- .70 is in subnet **.64** (.64–.127).
- .130 is in subnet **.128** (.128–.191). Different networks → traffic must go through a router (default gateway).

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**Q29.** Two PCs: 10.0.0.33 /27 and 10.0.0.60 /27. Same subnet?

**Answer**

**No.**

Block size 32 → subnets .0–.31, .32–.63, .64–.95...

- .33 is in **.32** (.32–.63).
- .60 is also in **.32** (.32–.63).

Wait — recheck: both are between 32 and 63! So they ARE in the same subnet **.32**. **Yes, same subnet — they can talk directly.**

(Lesson: always double-check by finding the block for BOTH addresses. Don't guess!)

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**Q30.** A device is configured 192.168.5.31 /27. What's wrong?

### Answer

**.31 is a broadcast address — you can't assign it to a host.**

Block size 32 → subnets .0–.31, .32–.63... **.31** is the **broadcast** of the first subnet (.0–.31). Assigning it will break connectivity.

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## SECTION 4 — "How Many Subnets / Hosts Do I Need?" (Design)

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(These are the word problems. Read carefully!)

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**Q31.** You have **192.168.1.0 /24** and need **4 equal subnets**. What mask and how many hosts each?

### Answer

**Mask /26 (255.255.255.192), 62 hosts each.**

- 4 subnets =  $2^2$ , so **borrow 2 bits**:  $/24 + 2 = /26$ .
- Host bits = 6 →  $2^6 - 2 = \mathbf{62}$  hosts.
- Subnets: .0, .64, .128, .192.

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**Q32.** You have **192.168.1.0 /24** and need **at least 50 hosts per subnet**, as many subnets as possible. What mask?

### Answer

**Mask /26 (255.255.255.192).**

- Need 50 hosts → find host bits:  $2^6 - 2 = 62 \geq 50$  ( $2^5 - 2 = 30$  is too small).
- 6 host bits → mask =  $32 - 6 = /26$ .
- Gives 4 subnets of 62 hosts each.

**Q33.** You need **at least 500 hosts** in one subnet. What is the smallest mask (largest CIDR number) that works?

**Answer**

**/23 (255.255.254.0).**

- $2^9 - 2 = 510 \geq 500$  ( $2^8 - 2 = 254$  too small).
- 9 host bits  $\rightarrow$  mask =  $32 - 9 =$  **/23**.

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**Q34.** You need **exactly 6 subnets** from **172.16.0.0 /16**, each as large as possible. What mask?

**Answer**

**/19 (255.255.224.0).**

- 6 subnets  $\rightarrow$  need  $2^3 = 8$  ( $2^2 = 4$  too few)  $\rightarrow$  **borrow 3 bits**.
- $/16 + 3 =$  **/19**. Host bits = 13  $\rightarrow 2^{13} - 2 =$  **8190** hosts each.

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**Q35.** How many **/30** subnets can you make from a single **/24**?

**Answer**

**64.**

From **/24** to **/30** you borrow 6 bits ( $30 - 24$ ).  $2^6 =$  **64** subnets. (Each **/30** has 2 usable hosts — great for point-to-point links.)

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**Q36.** How many **/26** subnets fit in a **/22**?

**Answer**

**16.**

$26 - 22 = 4$  borrowed bits  $\rightarrow 2^4 =$  **16** subnets.

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**Q37.** A company needs 200 hosts in one LAN. Will a **/25** work? What about a single **/24**?

**Answer**

**/25 = No. /24 = Yes.**

- /25 →  $2^7 - 2 = 126$  hosts (too small for 200).

- /24 →  $2^8 - 2 = 254$  hosts (fits 200).

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**Q38.** You need a subnet for a **point-to-point link between two routers** with no wasted addresses. What mask?

**Answer**

**/30 (or /31 for zero waste).**

- /30 gives exactly 2 usable hosts — one per router. The standard choice.

- /31 (RFC 3021) gives 2 usable with no network/broadcast waste, used on point-to-point links by advanced setups.

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**Q39.** From **10.0.0.0 /24**, you subnet into **/28s**. What is the **network address of the 5th subnet**?

**Answer**

**10.0.0.64**

Block size = 16. Subnets: 1st = .0, 2nd = .16, 3rd = .32, 4th = .48, **5th = .64.**

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**Q40.** From **192.168.1.0 /24** subnetted into **/27s**, what is the **broadcast of the 3rd subnet**?

**Answer**

**192.168.1.95**

Block size = 32. Subnets: 1st .0-.31, 2nd .32-.63, **3rd .64-.95**. Broadcast of 3rd = **.95.**

## SECTION 5 — VLSM (Variable Length Subnet Masks)

(Different-sized subnets from one block. The GOLDEN RULE: always assign the biggest subnet first!)

**Q41.** From **192.168.1.0 /24**, create subnets for: Sales (100 hosts), IT (50 hosts), HR (25 hosts), and a WAN link (2 hosts). Give the subnet + mask for each.

### Answer

### Assign biggest first:

| Group | Hosts | Mask      | Subnet        | Range     | Broadcast |
|-------|-------|-----------|---------------|-----------|-----------|
| Sales | 100   | /25 (126) | 192.168.1.0   | .1-.126   | .127      |
| IT    | 50    | /26 (62)  | 192.168.1.128 | .129-.190 | .191      |
| HR    | 25    | /27 (30)  | 192.168.1.192 | .193-.222 | .223      |
| WAN   | 2     | /30 (2)   | 192.168.1.224 | .225-.226 | .227      |

**Steps:** Sales needs 100 → /25 (126). Start at .0, uses .0-.127. Next free = .128. IT needs 50 → /26 (62), uses .128-.191. Next free = .192. HR needs 25 → /27 (30), uses .192-.223. Next free = .224. WAN needs 2 → /30, uses .224-.227.

**Q42.** From **10.0.0.0 /24**, you need three subnets: A (60 hosts), B (12 hosts), C (2 hosts). Provide subnet + mask for each.

### Answer

| Group | Hosts | Mask     | Subnet    | Range   | Broadcast |
|-------|-------|----------|-----------|---------|-----------|
| A     | 60    | /26 (62) | 10.0.0.0  | .1-.62  | .63       |
| B     | 12    | /28 (14) | 10.0.0.64 | .65-.78 | .79       |
| C     | 2     | /30 (2)  | 10.0.0.80 | .81-.82 | .83       |

A: 60 → /26 (62), .0-.63. Next = .64. B: 12 → /28 (14), .64-.79. Next = .80. C: 2 → /30, .80-.83.

**Q43.** In Q41, what goes wrong if you assign the **WAN /30 first** at 192.168.1.0, then try to fit Sales?

**Answer**

**You waste space and may not fit Sales cleanly (fragmentation).**

If WAN takes .0-.3, Sales (/25 = 126 hosts) can't start until a 128-aligned boundary (.128). You'd push Sales to .128-.255, leaving .4-.127 chopped up and possibly unusable for the other groups. **That's why VLSM's golden rule is: biggest subnet first.**

**Q44.** From 172.16.0.0 /22, allocate: Building1 (500 hosts), Building2 (250 hosts), Building3 (100 hosts). Provide subnet + mask for each.

**Answer**

| Group | Hosts | Mask      | Subnet     | Range                   | Broadcast    |
|-------|-------|-----------|------------|-------------------------|--------------|
| Bldg1 | 500   | /23 (510) | 172.16.0.0 | 172.16.0.1-172.16.1.255 | 172.16.1.255 |
| Bldg2 | 250   | /24 (254) | 172.16.2.0 | 172.16.2.1-172.16.2.255 | 172.16.2.255 |
| Bldg3 | 100   | /25 (126) | 172.16.3.0 | 172.16.3.1-172.16.3.127 | 172.16.3.127 |

Bldg1: 500 → /23 (510), uses 172.16.0.0-172.16.1.255. Next = 172.16.2.0. Bldg2: 250 → /24 (254), uses 172.16.2.0-.255. Next = 172.16.3.0. Bldg3: 100 → /25 (126), uses 172.16.3.0-.127. (All fits inside the /22!)

**Q45.** You subnetted 192.168.4.0 /24 with VLSM and used up to 192.168.4.223. A new team needs **20 hosts**. What subnet + mask do you give them, and does it fit?

**Answer**

**192.168.4.224 /27, and yes it fits exactly.**

- 20 hosts → /27 (30 usable). Next free address is .224.
- /27 block size 32 → uses .224-.255 (broadcast .255). Fits perfectly in the remaining space.

## SECTION 6 — Rapid-Fire Mixed Drills (Exam Simulation)

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(No hints. Do these on a timer — target under 60 seconds each.)

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**Q46.** 192.168.3.194 /26 → network, broadcast, first, last?

### Answer

Block 64. Subnets .0/.64/.128/.192. 194 is in **.192** (.192–.255).

- Network **192.168.3.192**, Broadcast **192.168.3.255**, First **.193**, Last **.254**.

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**Q47.** 10.20.30.40 /28 → network, broadcast, first, last?

### Answer

Block 16. 40 is between 32 and 47.

- Network **10.20.30.32**, Broadcast **10.20.30.47**, First **.33**, Last **.46**.

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**Q48.** 172.31.100.100 /26 → network, broadcast?

### Answer

Block 64. 100 is between 64 and 127.

- Network **172.31.100.64**, Broadcast **172.31.100.127**.

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**Q49.** 192.168.15.15 /29 → network, broadcast, first, last?

### Answer

Block 8. 15 is between 8 and 15.

- Network **192.168.15.8**, Broadcast **192.168.15.15**, First **.9**, Last **.14**.
- (Note: .15 is the broadcast, so the device .15 is NOT a usable host!)

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**Q50.** 10.0.5.130 /23 → network, broadcast? (Third-octet subnetting!)

**Answer**

/23 mask = 255.255.**254**.0. Block size in 3rd octet =  $256 - 254 = 2$ . Count by 2 in octet 3: 0, 2, 4, 6... 5 is between **4 and 5**.

- Network **10.0.4.0**, Broadcast **10.0.5.255**.

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**Q51.** 192.168.1.222 /27 → is it a host, network, or broadcast?

**Answer**

**Usable host.**

Block 32. Subnet containing 222 is **.192** (.192-.223). Network = .192, Broadcast = .223. Since 222 is between .193 and .222, it's a **usable host**.

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**Q52.** 172.16.14.0 /23 → how many usable hosts, and what's the broadcast?

**Answer**

**510 hosts; broadcast 172.16.15.255.**

/23 → host bits 9 →  $2^9 - 2 = 510$ . Block size 2 in 3rd octet → 14 and 15 belong together (14-15), so range is 172.16.14.0-172.16.15.255, broadcast **172.16.15.255**.

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**Q53.** 10.10.10.100 /25 → which subnet (network address)?

**Answer**

**10.10.10.0**

Block 128. Subnets: .0 (.0-.127) and .128 (.128-.255). 100 is in **.0**. Network = **10.10.10.0**.

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**Q54.** 192.168.100.100 /30 → network, broadcast, both usable hosts?

**Answer**

Block 4. Count by 4: ...96, 100, 104. 100 is a boundary.



- Network **192.168.100.100**, Broadcast **192.168.100.103**, Hosts **.101 and .102**.

**Q55.** You see **255.255.255.252**. How many of these subnets fit in a **/24**, and what are they good for?

#### Answer

**64 subnets; good for point-to-point router links.**

255.255.255.252 = **/30**. From /24 to /30 you borrow 6 bits →  $2^6 = 64$  subnets, each with **2 usable hosts**. Perfect for router-to-router (WAN) links where you only need one address on each end.

## SECTION 7 — Boss Level: Full Scenario Problems

(These mimic the harder simulation-style exam questions. Take your time.)

**Q56.** A router interface is configured with **192.168.1.65 /26**. A host on that LAN is set to **192.168.1.130 /26** with gateway **192.168.1.65**. The host can't reach the internet. What's the problem and the fix?

#### Answer

**Problem: the host and its gateway are in different subnets.**

- Gateway .65 /26 is in subnet **.64** (.64-.127).
- Host .130 /26 is in subnet **.128** (.128-.191).

The host can't reach a gateway that isn't on its own subnet, so it can't route out. **Fix:** give the host an address in .64-.127 (e.g., 192.168.1.66), OR move the gateway into the host's subnet. They must share the same subnet.

**Q57.** You're given **192.168.50.0 /24** and must support **6 subnets**, each needing **25 usable hosts**. Give the mask and list all 6 network addresses.

#### Answer

### Mask /27 (255.255.255.224).

- 25 hosts → /27 gives 30 usable .
- /27 also gives  $2^3 = 8$  subnets ( $\geq 6$  needed) .
- Block size 32 → networks: **.0, .32, .64, .96, .128, .160** (first 6), plus .192, .224 spare.

**Q58.** 10.1.1.0 /24 needs these, biggest-first (VLSM): Subnet A 120 hosts, B 60 hosts, C 30 hosts, D 12 hosts, plus two /30 WAN links (E, F). List each subnet + mask + range.

### Answer

| Sub | Hosts | Mask      | Network    | Range     | Broadcast |
|-----|-------|-----------|------------|-----------|-----------|
| A   | 120   | /25 (126) | 10.1.1.0   | .1-.126   | .127      |
| B   | 60    | /26 (62)  | 10.1.1.128 | .129-.190 | .191      |
| C   | 30    | /27 (30)  | 10.1.1.192 | .193-.222 | .223      |
| D   | 12    | /28 (14)  | 10.1.1.224 | .225-.238 | .239      |
| E   | 2     | /30 (2)   | 10.1.1.240 | .241-.242 | .243      |
| F   | 2     | /30 (2)   | 10.1.1.244 | .245-.246 | .247      |

Everything fits inside the /24 with room to spare (.248-.255 free).

**Q59.** Summarize (aggregate) these four networks into ONE route: 172.16.0.0/24, 172.16.1.0/24, 172.16.2.0/24, 172.16.3.0/24.

### Answer

#### 172.16.0.0 /22

The four /24s (.0, .1, .2, .3 in the 3rd octet) share the top 22 bits. In binary the 3rd octet: 00000000, 00000001, 00000010, 00000011 — the first **6 bits** are identical, and  $16 + 6 = 22$ . So the summary is **172.16.0.0/22** (covers .0 through .3). This is **route summarization**, which shrinks routing tables.

**Q60.** Summarize 192.168.8.0/24 through 192.168.15.0/24 into one route.

### Answer

**192.168.8.0 /21**

Third octet values 8-15 in binary: 00001000 ... 00001111. The first **5 bits** (00001) are common →  $16 + 5 = 21$ . Summary = **192.168.8.0/21** (covers .8 through .15).

## The One-Page Speed Method (Print This!)

When you see any "IP + CIDR" question, do this in your head, every time:

```
STEP 1: Which octet is "interesting"?
        /1-8 → 1st octet
        /9-16 → 2nd octet
        /17-24 → 3rd octet
        /25-32 → 4th octet

STEP 2: Block size = 256 - (mask value in that octet)

STEP 3: Count 0, block, 2×block... until you pass the IP.
        The one at/just-below = NETWORK.

STEP 4: Broadcast = next network - 1
        First host = network + 1
        Last host = broadcast - 1

STEP 5: Usable hosts =  $2^{(32-CIDR)} - 2$ 
```

## Final Tips

1. **Memorize the cheat table** (/24-/30 with block sizes and host counts). It saves you 30 seconds every question.
2. **Always find the block for BOTH addresses** when asked "same subnet?" — don't eyeball it.
3. **Watch out for network & broadcast addresses** disguised as host answers (a favorite trick).
4. **VLSM golden rule:** biggest subnet first, every time.
5. **Do 10 problems a day.** Speed comes only from reps. In two weeks this will feel easy.

You've got this. Now go drill!

— End of Subnetting Drill Sheet —